

The Cloud

Introduction to the Cloud

The term cloud describes the internet when it is used to provide software or space for storing information, rather than relying on local storage.

- **Local Hardware:** When you use software stored directly on your computer's hard drive to work with files, you are using local hardware. Because there is no network connectivity required to transfer data, this method generally provides the best performance speed.
- **Cloud Computing:** When you use the cloud, you use the internet to access applications and files that are stored on remote servers housed in data centers. Examples include using web applications like Twitter or Prezi, or syncing local folders with online storage systems.

Hosted Applications Software

Web applications (web apps)—also known as **hosted applications**—are programs stored and run on online servers. The user interface is displayed to you through a standard web browser. This setup relies on the client-server model.

Performance Factors

Because hosted applications are only available through an active internet connection, their performance is heavily dependent on **latency** (delay) and **bandwidth** (data capacity). Low bandwidth or high latency can result in:

- Slow application loading times.
- A noticeable delay between user input and browser activity, making the application seem **unresponsive**.

To minimize these performance drops, hosted software usually offers fewer complex features than locally installed software. Processing less data keeps the user experience smoother.

Collaboration

Because hosted software operates on a client-server model, multiple users can access the exact same application or file simultaneously. This enables real-time collaboration, often supported by integrated chat functions and tracking tools to add or resolve edits and comments.

Benefits and Drawbacks of Hosted Applications

Benefits	Drawbacks
Accessible from many types of connected devices.	Only accessible when you have an active internet connection.
Often includes integrated online data storage to save files online.	User experience can be poorly affected by high latency or low bandwidth.
Features powerful built-in tools that enable group collaboration.	Generally has fewer advanced features compared to local software.

Online Data Storage

Just like hosted applications, online data storage allows you to access files from any device in any location, provided you have an internet connection.

Benefits and Drawbacks of Online Data Storage

Benefits	Drawbacks
Files are frequently backed up automatically by the provider.	Potential data protection issues if data is stored on servers in another country.
The storage is scalable (it can expand or shrink based on demand).	Security is entirely dependent on the online storage provider.
Sharing data with other users is much easier.	File transfer speeds are significantly slower than local storage.
Many providers offer unlimited storage packages, reducing local hardware needs.	Only accessible when you have an internet connection.
Enables real-time document collaboration.	Some file formats cannot be opened online without downloading them first.
Frees up local storage space for local applications and files.	Users often have to pay subscription fees for additional storage space.

The Impact of the Cloud

On Technology and the Digital Divide

The growth of cloud services means devices no longer require massive internal storage drives or ultra-fast processors. This has led to the manufacturing of inexpensive computers like Chromebooks and netbooks. Lower hardware costs allow more people to access modern technology, helping to reduce the digital divide.

On Organizations

Cloud services provide businesses with highly scalable infrastructure that adjusts to their changing needs. It eliminates the massive overhead expenses associated with purchasing, housing, and maintaining physical in-house servers.

On Individuals (Online Services)

The cloud underpins many essential digital services used daily by individuals, including:

- Shopping and online auction platforms
- Travel, leisure, and entertainment booking systems
- Digital banking
- Virtual Learning Environments (VLEs) and Massive Open Online Courses (MOOCs)
- Online gaming
- Digital news and information portals
- On-demand entertainment services (streaming television and radio)

On Online Communities

Cloud infrastructure connects people globally by hosting interactive communities such as:

- Social networking sites
- Online workspaces for remote employment
- User-generated reference portals, including wikis and forums
- User-generated content platforms, such as blogs and video-sharing sites
- Social bookmarking sites

Vocabulary Reference

- **The Cloud:** The internet when it is used to provide software or space for storing information, rather than relying on local storage.
- **Application Loading Time:** The total time it takes for an application to open and become ready for use.
- **Unresponsive:** Not reacting to user input or commands, often caused by lag.
- **Edit:** A modification or change made to a document.
- **Scalable:** The capacity of a system (like storage or processing) to grow or shrink flexibly to perfectly meet varying demands.

- **Data Protection Issues:** Security problems or compliance risks associated with maintaining the privacy and safety of information.
 - **Netbook:** A compact computer designed primarily for internet access that lacks large internal storage or high-speed processing components.
-

Review Questions

1. **Multiple Choice:** Which one of these options is used to store and run hosted applications?
 - A) Laptop
 - B) Smartwatch
 - C) Server
 - D) Tablet
 2. **Short Answer:** State three distinct benefits of utilizing online data storage.
 3. **Short Answer:** Identify three types of everyday online services made possible by cloud-based services.
 4. **Definition:** In your own words, state what is meant by the term "the cloud".
-